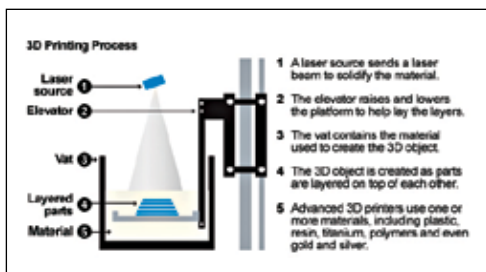


Basic principles of a 3D printer

Forget everything you think you know about printing, manufacturing, assembling and designing objects. The world of 3D printing has developed in such a way that it disrupts every aspect of conventional product manufacturing. The underlying principles of 3D printing and the printers themselves are moving towards a unified state of turn-key manufacturing limited only by our imagination.

The most essential aspect that needs to be understood about 3D printing or ‘additive printing’ as a personal user tool, is its process. Although there are different ways by which the varied models of 3D printers operate, they’re all based on a simple premise. As the term ‘additive printing’ suggests, 3D printers work by “adding” layers of print material together to create an object. Converting a software-based design into distinct 2D layers or slices, which are “printed” and bonded to each other in order to create a 3D product is the primary method of operation of any 3D printer.



A simple workflow. It doesn't look that complicated, does it?

Imagine placing a dot of glue on a piece of paper. Now imagine adding layer upon layer of glue on that dot in a precise and adhesive manner. That dot would grow off the two-dimensional page into the third dimension and become a cylinder with the diameter of the original dot. Another example: a single sheet of paper would be two-dimensional but a stack of sheets would make it three-dimensional. In the exact same way, 3D printers create 3D objects by printing layer upon layer of a variety of materials to achieve a three-dimensional product.

The overall workflow of any 3D printer is oriented towards achieving the goal of converting a 3D design created using software into a hardcopy version. While the methods used by different printer models vary, they're all based on the same type of workflow. Let's look at this workflow and then delve deeper into the world of 3D Printing by exploring the different types of 3D printers available in the market.